Capital Allocation III

When you have to estimate the inputs

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Today

In the slides

- Why estimation error makes the optimizer swing
- Which input does the damage: means, volatilities or correlations
- When the optimizer still wins, and shrinkage as the way in between

In the notebook, you will

- Race mean-variance against 1/N, and check which months went into each estimate
- Run six rules through the race
- Bootstrap the weights, one input at a time

Homework: audit the book you built last class

In sample, the optimizer always wins

six factors, 1980–2026	Sharpe ratio, in sample
mean-variance, $\Sigma^{-1}\mu$	1.17
1/N	1.03
the best single factor (the market)	0.57

- It had to win. $\Sigma^{-1}\mu$ *is* the maximum in this sample. That is algebra, not evidence.
- The real question: what does the formula do with inputs that are slightly wrong?

Optimization is in-sample fitting, and everything from *Backtesting* applies to it

A small error in the means does not stay small

$$W^\star = \tfrac{1}{\gamma}\,\Sigma^{-1}\mu \qquad \Longrightarrow \qquad \Delta W = \tfrac{1}{\gamma}\,\Sigma^{-1}\,\Delta\mu$$

The error is multiplied by the inverse of the covariance matrix.

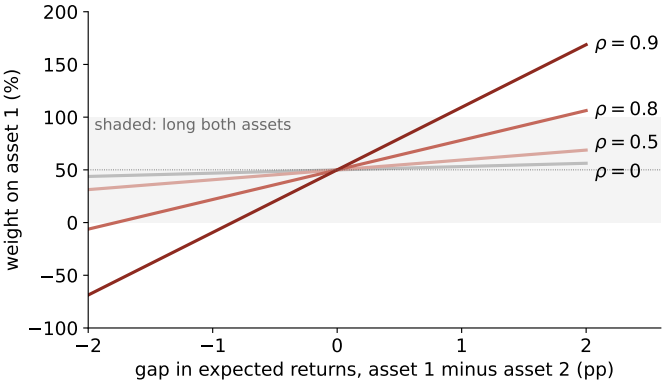
Two assets, same volatility, weights adding to one:

$$w_1 \;=\; \frac{1}{2} \;+\; \frac{\text{gap}}{4 \times \text{average mean}} \times \frac{1+\rho}{1-\rho}$$

correlation ρ	0	0.5	0.8	0.9	0.95
amplifier $(1+\rho)/(1-\rho)$	1	3	9	19	39
weight on asset 1, 1pp gap, average mean 8%	53%	59%	78%	109%	172%

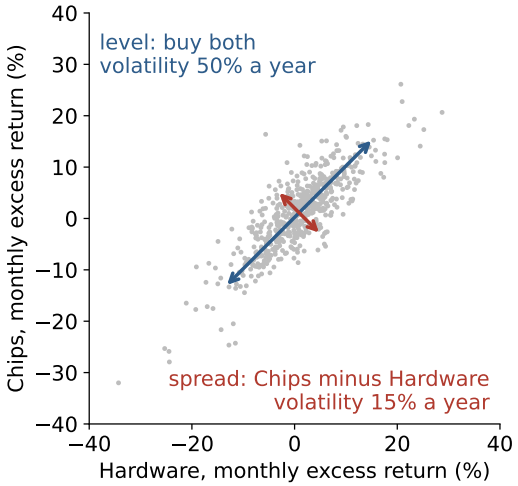
Derivation

One percentage point, four different portfolios



- Zero correlation: a 1pp gap barely matters
- At $\rho = 0.9$: 50/50 becomes $109/-9$
- Two good assets, and it shorts one
- Nobody can tell a 1pp gap in expected returns from zero

Why: correlation makes the spread look almost riskless

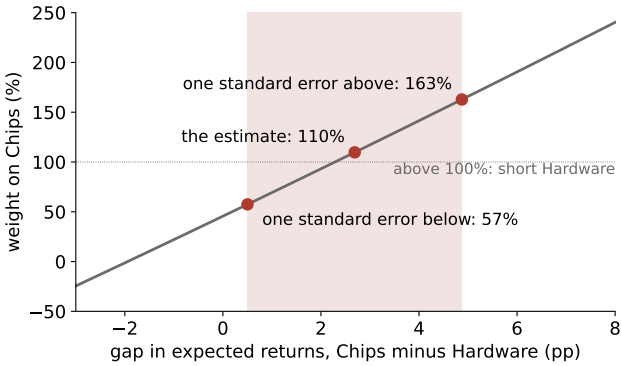


Any pair is two bets: a level and a spread.

$$\frac{\text{Var(level)}}{\text{Var(spread)}} = \frac{1+\rho}{1-\rho} = \frac{50\%^2}{15\%^2} = 11$$

- The optimizer sizes every bet by mean over variance
- The spread has a tiny variance, so a tiny gap earns a huge position
- The amplifier is that ratio of variances

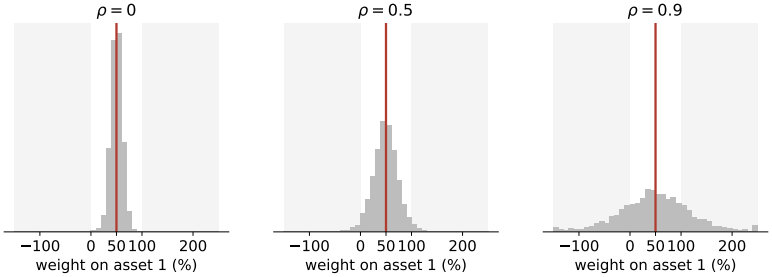
A real pair: Hardware and Chips, $\rho = 0.84$



1980–2026	mean ± s.e.
Hardware	10.6% ± 3.8%
Chips	13.3% ± 3.9%
gap	2.7% ± 2.2%

- 46 years, and the gap is not distinguishable from zero
- One standard error is the difference between 57% and 163% in Chips
- At the estimate it shorts Hardware, a sector with a Sharpe ratio of 0.41

Same truth, same 46 years, three correlations



Both assets: true mean 8%, volatility 16%. The right answer is always 50/50 (red). Shaded: a short position.

- 90% of histories: 32% to 69% at $\rho = 0$; −65% to 165% at $\rho = 0.9$, short in 44% of them
- In fairness, correlation also helps: the gap is better estimated, its error falls with $\sqrt{1-\rho}$
- But the amplifier grows with $1/(1-\rho)$. Net, the weight's error grows with $(1+\rho)/\sqrt{1-\rho}$

The formula has three inputs, each with a standard error

input	standard error	example, per year	error, % of estimate
expected return	$\sigma/\sqrt{T}$	HML mean: 2.8% $\pm$ 1.6%	57%
volatility	$\sigma/\sqrt{2T}$	HML volatility: 10.9% $\pm$ 0.3%	3%
correlation	$(1-\rho^2)/\sqrt{T}$	market and HML: -0.21 ± 0.04	19%

- T is the number of observations in the time series: here 558 months. σ is the volatility per observation.
- Daily data raises T, and shrinks the error on volatilities and correlations. The error on a yearly mean does not move: only more years help.
- With N assets: N means, N volatilities, $N(N-1)/2$ correlations
- What matters is not the size of the error, but what it does to the portfolio.

Poll

Six factors, 46 years of data. You re-estimate one kind of input and leave the other two alone.

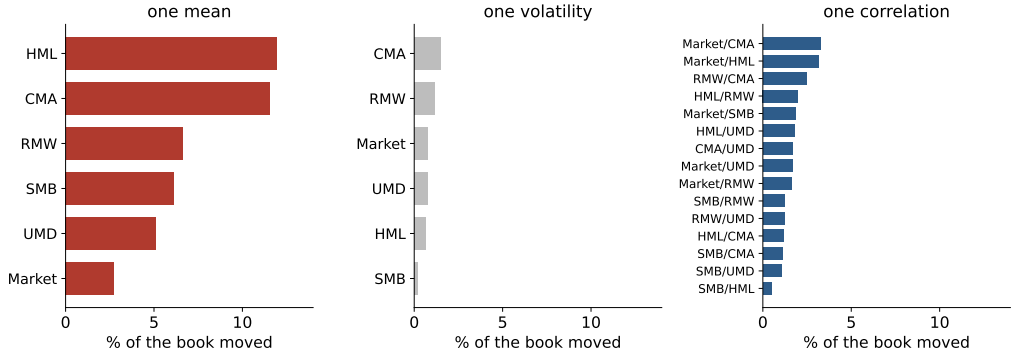
Which one moves your portfolio the most?

A the means

B the volatilities

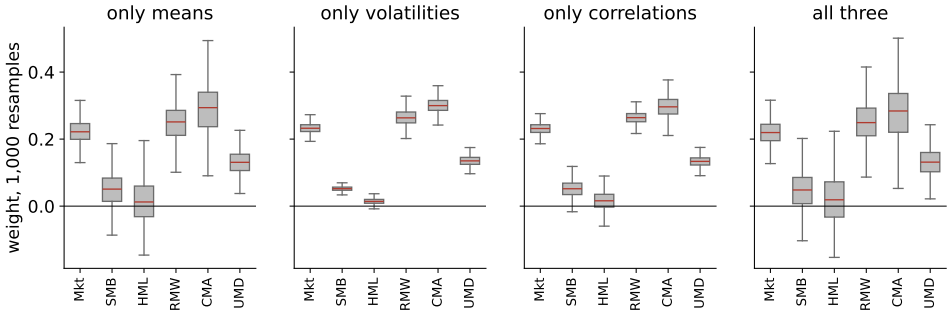
C the correlations

Exercise 1: move one number by one standard error



- Every bar: change one input, hold the other 26 fixed, count how much of the book changes hands
- Largest move: a mean 12%, a correlation 3.3%, a volatility 1.5%
- But standard errors do not arrive one at a time, and there are 15 correlations

Exercise 2: move them all, as history would



- Resample the 558 months, 1,000 times. Each draw gives new means, volatilities and correlations.
- Feed in one kind at a time, the other two at their full-sample values. Then all three.
- Variance of the weights: 79% means, 14% correlations, 7% volatilities

The bootstrap, in a picture

Who is to blame depends on what you hold

	variance of the weights			Sharpe ratio lost			Sharpe lost,
	means	vols	corrs	means	vols	corrs	ten years of data
six factors	79%	7%	14%	3.8%	0.5%	0.6%	18%
eight sectors	88%	6%	6%	10.1%	0.4%	0.5%	34%
five asset classes	96%	2%	2%	10.3%	0.3%	0.2%	36%
29 long-shorts	35%	8%	57%	5.3%	1.2%	8.8%	27%

- Volatilities are never the problem
- A few assets: the means. 29 strategies have 406 correlations, some near 0.97: they overtake the means.
- Sharpe lost as a share of the best, using all the data unless stated

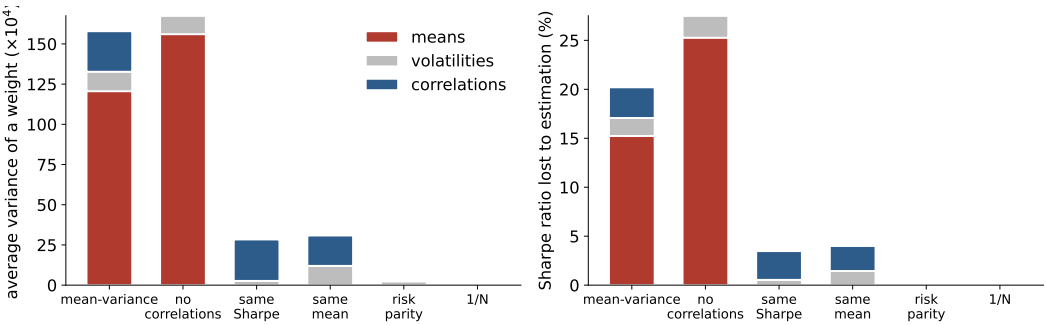
Careful: this world takes the sample means as the truth. So it overstates how different the assets are. We come back to this.

Each rule replaces an estimate with a constant

rule	weights	the constant	still estimates		
mean-variance	$\Sigma^{-1}\mu$	none	means	vols	corrs
no correlations	μ_i/σ_i^2	every correlation is zero	means	vols	
same Sharpe	$\Sigma^{-1}\sigma$	every Sharpe ratio is equal		vols	corrs
same mean	$\Sigma^{-1}\mathbf{1}$	every expected return is equal		vols	corrs
risk parity	$1/\sigma_i$	same Sharpe, no correlations		vols	
1/N	$1/N$	same Sharpe, no correlations, same vol			

- "Same mean" is minimum-variance. Risk parity is $w_i\sigma_i \propto AR_i$ from *Capital Allocation II*, with the ratios set equal.
- None of these is simply better. Each is a claim about your assets.

Change the rule, change what you are exposed to



Six factors, each history ten years long. Each input moved separately.

- Stop estimating the means and the exposure collapses: Sharpe lost 18% to 3%. With all 46 years: 5% to 1%.
- Ignoring correlations does not help. The noise in the means is still there, with no hedge against it.
- 1/N has no exposure at all. Is that free? Only if its constant is right.

Now: the notebook

Notebook: Capital Allocation III — the rules, out of sample

1. **"Is this out of sample?"** Race mean-variance against 1/N with AI, then read which months went into each estimate.
2. **The menu.** The six rules, the race, and the bootstrap one input at a time.

Bring back: which rule won the race, and would you bet that it wins on *any* set of assets?

Stop at **Back to the slides**; the challenge after it is homework.

1/N won. One index would have reversed it.

six factors, expanding window, ten-year burn-in	Sharpe ratio, out of sample
1/N	0.92
mean-variance	0.83
same Sharpe	0.83
same mean	0.77
risk parity	0.76
mean-variance, estimate includes the month it earns	1.03

- DeMiguel, Garlappi and Uppal (2009): the formula loses to dividing by the number of assets
- Written with `F.iloc[:t+1]` instead of `F.iloc[:t]`, the conclusion flips. Your AI may not warn you.
- The gap is 0.09 ± 0.10. It is not a law of nature, so run the same race on other assets.

Same race, other assets: the winner changes

ten-year burn-in, before costs	period	mean-variance	1/N	1/N minus mean-variance
29 long-shorts	1980–2000	2.24	0.98	-1.25 ± 0.39
six factors	1980–2026	0.83	0.92	$+0.09 \pm 0.10$
eight sectors	1980–2026	0.46	0.64	$+0.18 \pm 0.17$
five asset classes	1963–2016	0.39	0.50	$+0.11 \pm 0.13$

- "1/N beats the optimizer" is not a result. It is a result about particular assets.
- So what do the assets have to look like?

Careful with the 29: they were chosen later because they worked in this very period, and the optimizer trades them heavily. Treat 2.24 as an upper bound. The gaps in the last three rows are within about one standard error.

The race, in one formula

The optimizer sees N independent bets, and from *Capital Allocation II* their Sharpe ratios add in squares.

- Each bet also carries a fake Sharpe ratio of about $\pm 1/\sqrt{Y}$, with Y the years of data and Sharpe ratios per year
- The squares of the fakes add up too: N/Y

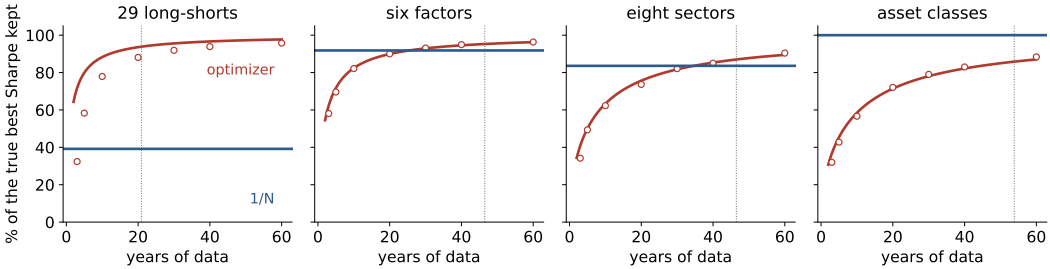
$$\underbrace{SR^2_{\text{in sample}}}_{\text{what you see}} \approx \underbrace{SR^2}_{\text{real differences}} + \underbrace{N/Y}_{\text{estimation error}} \qquad \Longrightarrow \qquad \frac{SR_{\text{you earn}}}{SR} \approx \frac{1}{\sqrt{1 + \dfrac{N}{Y\, SR^2}}}$$

A race between how different your assets really are and how noisy your estimates are

- Correlation enters through SR: for the same assets, high ρ means a lower SR, so noise matters more

Derivation

The optimizer learns, 1/N does not, and the crossing moves



A world we control, built from each asset set. Line: the formula. Dots: simulation, estimating everything. Dotted: the years we actually have.

- 29 long-shorts: differences so large the optimizer wins almost at once
- Factors and sectors: it takes decades. Asset classes: 1/N is already about as good as it gets.

The simulated world

Real differences, or noise?

	Sharpe ratio, in sample		extra SR^2 beyond 1/N		noise	years to
	1/N	best	seen	noise, $(N-1)/Y$	share	beat 1/N
29 long-shorts	1.25	3.40	10.01	1.34	13%	under 1
six factors	1.03	1.17	0.31	0.11	35%	25
eight sectors	0.61	0.83	0.31	0.15	48%	35
five asset classes	0.51	0.56	0.06	0.07	128%	never

- It lines up with the race, whose average window was 28 years: under 1 year is an easy win, 25 is a coin flip, 35 is a loss
- Asset classes: all of the extra Sharpe ratio you see is what noise alone would produce

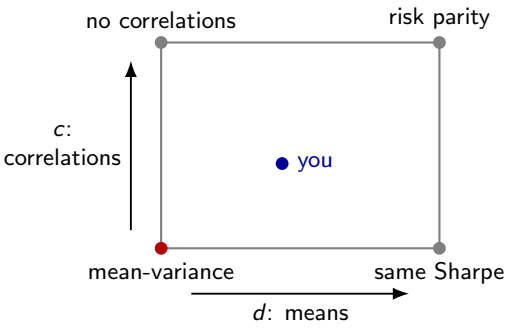
Similar assets: error dominates, assume a constant. Very different assets: a constant throws the signal away.

Which constant? It has to fit the assets too

	range, in sample		Sharpe ratio, out of sample			
	means	Sharpes	same mean	same Sharpe	mean-variance	1/N
five asset classes	2.5% to 8.1%	0.25 to 0.41	0.45	0.55	0.39	0.50
eight sectors	7.3% to 11.9%	0.30 to 0.76	0.44	0.71	0.46	0.64
six factors	0.7% to 8.9%	0.07 to 0.57	0.77	0.83	0.83	0.92
29 long-shorts	−6.4% to 19.9%	−0.57 to 1.36	1.07	1.08	2.24	0.98

- Same mean on stocks and bonds is ill-advised: it puts 82% in foreign government bonds, for having low volatility
- Bonds and emerging markets do not earn the same return. They may earn about the same Sharpe ratio.
- Sectors have similar means, yet same mean does not rescue them: 0.44 against 0.46. Same Sharpe does.
- On the 29, either constant throws away half the Sharpe ratio. Those assets really are different.

Shrinkage: the same choice, made continuous



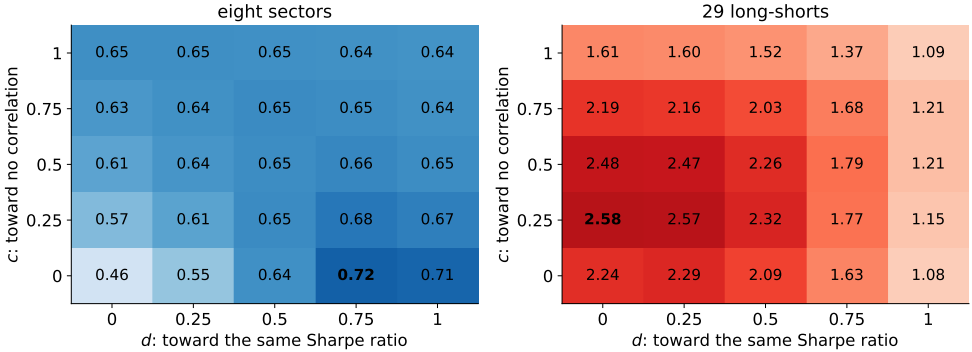
One dial per input:

$$s_i(d) = (1-d)\,\hat{s}_i + d\,\bar{s}$$

$$R(c) = (1-c)\,\hat{R} + c\,I$$

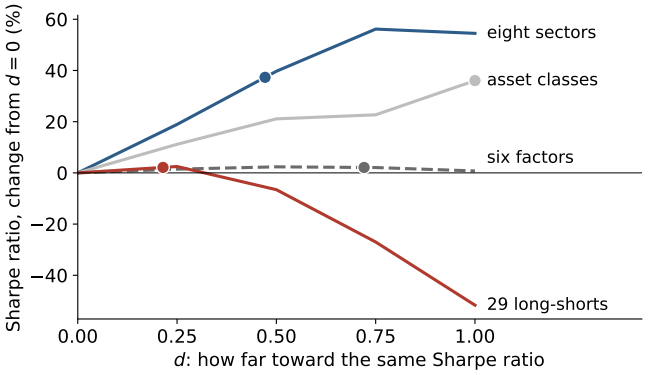
- The rules are the corners. Shrinkage lets you stand anywhere in between.
- Set all the volatilities equal too and risk parity becomes 1/N
- Long-only and position caps work the same way: a constraint is a prior

Same two dials, opposite answers



- Sectors: turn the means dial, 0.46 $\rightarrow$ 0.72. Their means are mostly noise.
- The 29: leave the means, turn the correlations a little, 2.24 $\rightarrow$ 2.58. Just as Exercise 2 said: 57% of their weight variance came from correlations.
- Out of sample, before costs. Picking the best cell afterward is itself in-sample fitting.

How far to turn the dial is the race again



$$d \approx \frac{\text{noise}}{\text{dispersion you see}}$$

in the Sharpe ratios across assets; the dots

- It can help or hurt: $+56\%$ on sectors, -52% on the 29
- The rule gets the direction and the rough size. On sectors it says 0.47, the best was 0.75.
- The logic of Black-Litterman: weigh the prior against the noise in the estimate

What to actually do

1. **Ask how different your assets really are.** Compare SR^2 with N/Y before you trust a formula, or a rule of thumb.
2. **Look at the correlations.** Above about 0.8, you are mostly trading a spread. Ask whether you believe the gap.
3. **Use data on what data can teach**: volatilities first. Correlations too, unless N is large.
4. **Turn dials, do not pick sides.** Similar assets: near the constant. Very different assets: near the estimates.
5. **Treat the dial as a parameter**: tune it without the future, test it, report once, and count costs.

Takeaways

1. The formula multiplies input errors by Σ^{-1}. For two correlated assets the amplifier is $(1+\rho)/(1-\rho)$.
2. In a small book the means do the damage. In a large one the correlations can take over. Volatilities never do.
3. Each rule replaces an estimate with a constant, and is exposed only to what it still estimates.
4. Whether that helps is a race: real differences, SR^2, against estimation error, N/Y.
5. 1/N beats the optimizer on similar assets, and loses badly on very different ones.
6. Shrinkage is the rules made continuous, and how far to go depends on the assets.

Homework: audit the book. Throw the alphas away and see what happens, out of sample. Then find the right amount of shrinkage, and find out whether you could have known it.

Appendix

A1: Two assets

Same volatility σ, correlation ρ, means $\mu_1 = \bar{\mu} + g/2$ and $\mu_2 = \bar{\mu} - g/2$.

$$\Sigma^{-1}\mu \;\propto\; \begin{pmatrix} 1 & -\rho \\ -\rho & 1 \end{pmatrix} \begin{pmatrix} \mu_1 \\ \mu_2 \end{pmatrix} \;=\; \begin{pmatrix} \mu_1 - \rho\mu_2 \\ \mu_2 - \rho\mu_1 \end{pmatrix}$$

The two entries add to $2\bar{\mu}(1-\rho)$, and the first is $\bar{\mu}(1-\rho) + \frac{g}{2}(1+\rho)$. So

$$w_1 = \frac{1}{2} + \frac{g}{4\bar{\mu}} \cdot \frac{1+\rho}{1-\rho}$$

- Asset 2 is shorted when $\mu_2/\mu_1 < \rho$. With different volatilities, the same test on Sharpe ratios.
- With Y years of data, the error in g is $\sigma\sqrt{2(1-\rho)/Y}$, so the error in w_1 is $\dfrac{1+\rho}{4\,(\bar{\mu}/\sigma)}\sqrt{\dfrac{2}{Y(1-\rho)}}$. With $\bar{\mu}/\sigma = 0.5$, $Y = 46$: 10%, 22%, 62%.

Back

A2: The formula for the race

Whiten the assets: $z = \Sigma^{-1/2}\mu$, so $|z|^2 = SR^2$. The estimate of z is $z + u$, with u noise of variance $1/Y$ in each of N directions.

- In sample you see $|z + u|^2 \approx SR^2 + N/Y$
- Your portfolio points along $z + u$. Under the truth it earns $\dfrac{z'(z+u)}{|z+u|} \approx \dfrac{SR^2}{\sqrt{SR^2 + N/Y}}$
- Against 1/N, which keeps a share q of the true best, the optimizer wins once $Y > \dfrac{N}{SR^2} \cdot \dfrac{q^2}{1-q^2}$
- This covers errors in the means only. Errors in Σ add to it when N is large relative to the data: see the dots for the 29.

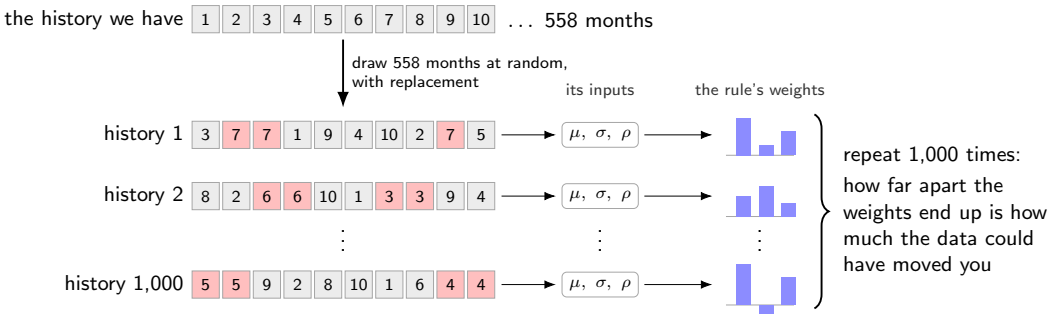
Back

A3: The simulated world

- **Covariances**: each asset set's sample Σ, with normal returns
- **1/N**: keeps its sample Sharpe ratio. No selection went into it, so it needs no correction.
- **The rest**: the extra SR^2 beyond 1/N, minus the $(N-1)/Y$ that noise alone would produce
- **Score**: the Sharpe ratio under the truth, over the true best. Mean of 300 histories.
- **The limit**: a stable world, correctly specified, and a noisy estimate of how much signal there is. Real markets move, so read the years as a lower bound.

Back

A4: The bootstrap, in a picture



red: a month drawn more than once. Others are left out. Each history is as long as the real one.

- Each made-up history is one way the past could have gone. The rule reads it as if it were the truth.
- Exercise 2 hands the rule one new input at a time from each history, say the new means, with the volatilities and correlations of the real one

Back

A5: How not to measure the cost

The obvious test: fit on one ten-year window, run on the next, compare with the best that window allowed.

	share of the "best" Sharpe ratio lost
weights estimated on the previous window	46%
weights that saw the whole sample, future included	19%

- A portfolio with no forecast error is charged 19% for forecast error
- The benchmark is an in-sample maximum, so everything loses to it

Before you trust a test, run it on something that cannot be wrong

A6: The same assets, two periods

six factors, five-year burn-in in each period	mean-variance	1/N	gap
1980–1999	1.26	1.54	+0.28
2000–2026	0.57	0.59	+0.01

- Even on one set of assets the gap moves with the period
- Each period restarts its own burn-in, so these are not the same months as the full-sample race

A7: Data, and what to be careful about

- **Six factors.** Ken French, 1980–2026. **Eight sectors.** Built from his 49 industries, equal-weighted, 1980–2026.
- **Asset classes.** US and world stocks, emerging markets, US and world government bonds, 1963–2016, over T-bills. A different and older sample, and it stops in 2016.
- **29 long-shorts.** 1980–2000, not hedged here. Published later because they worked, so selection flatters them.
- **The race.** Weights use only months before the one they earn. Everything is before costs, and mean-variance turns over far more than 1/N.
- **The bootstrap** assumes months are independent, and scores against the sample as if it were the truth. Volatility clusters and means move, so true uncertainty is, if anything, larger.